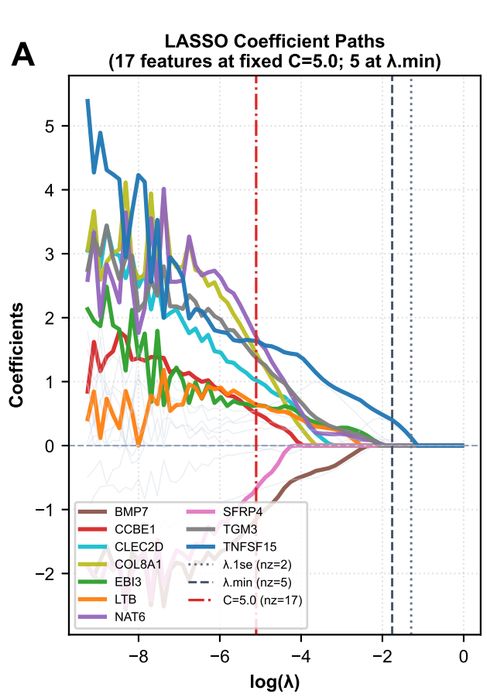
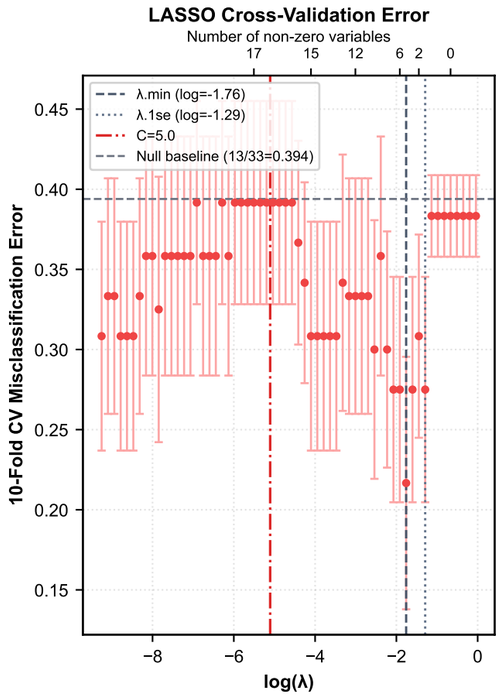
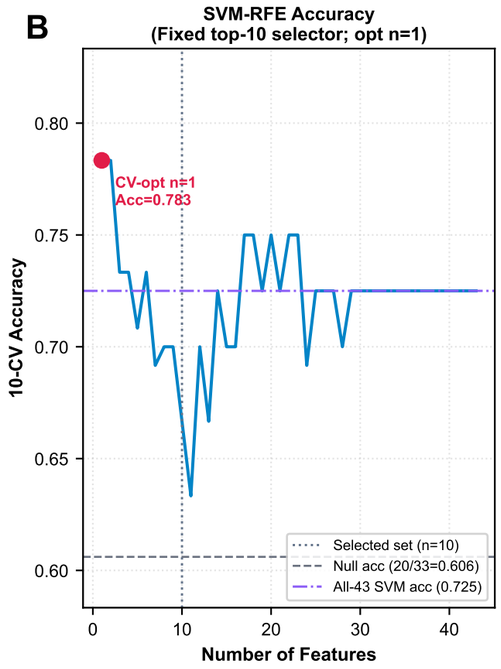
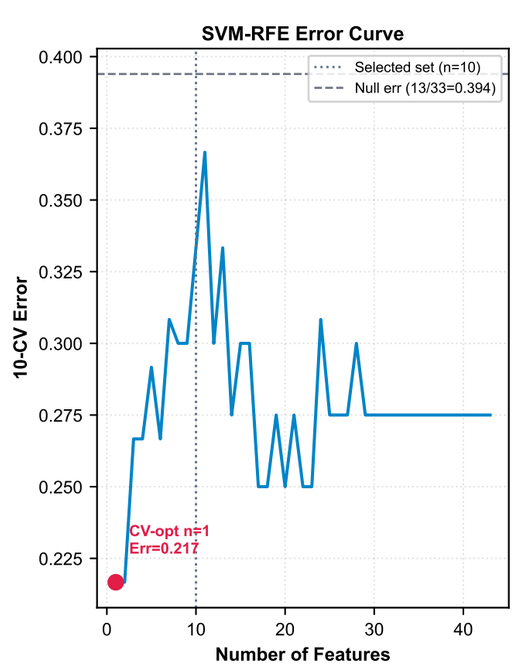
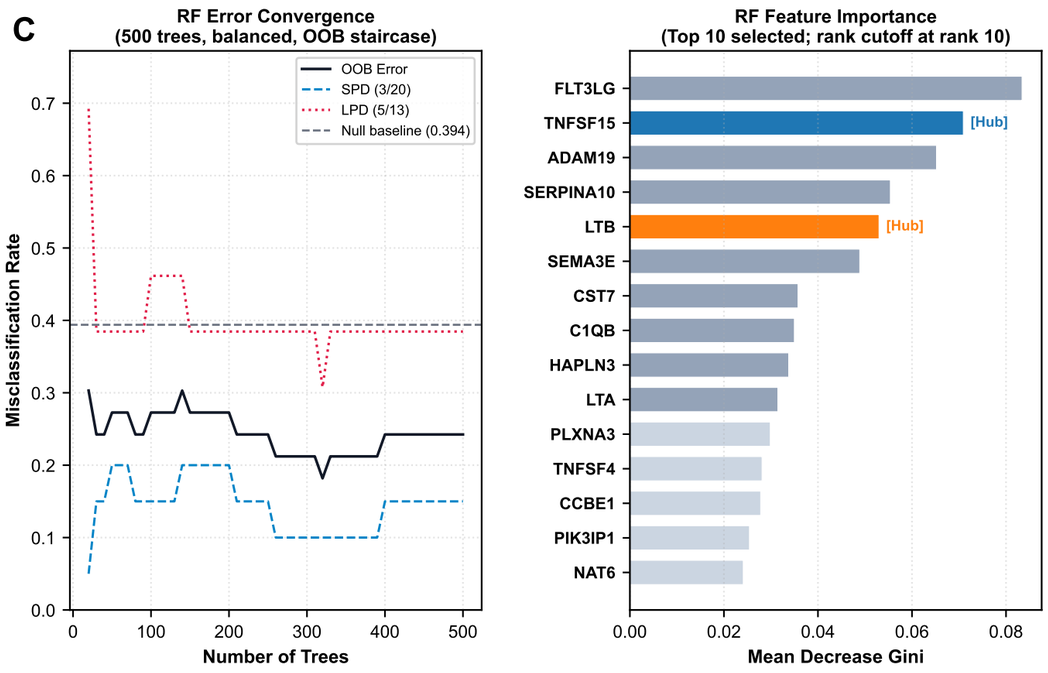
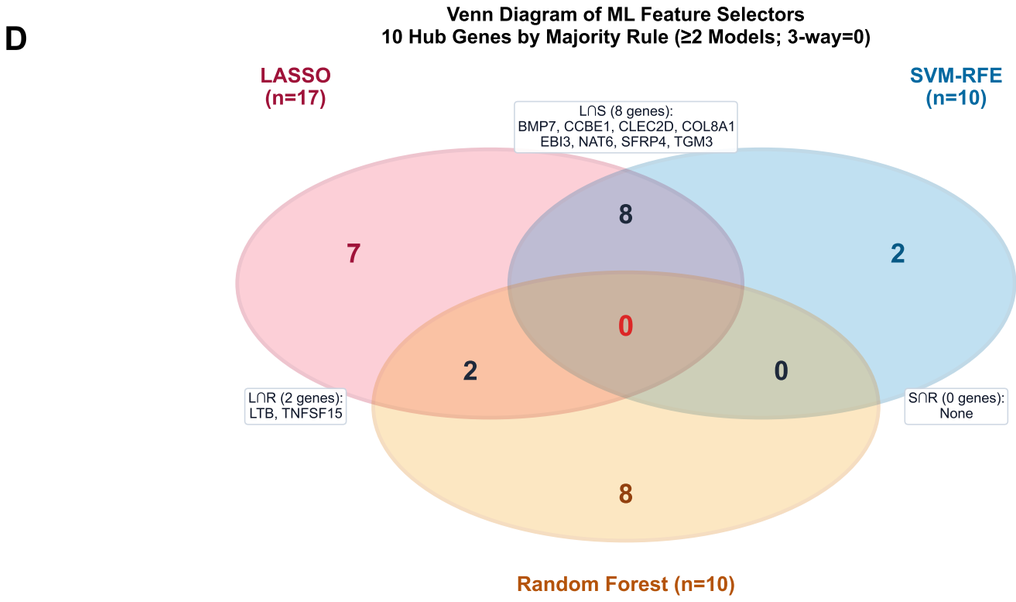


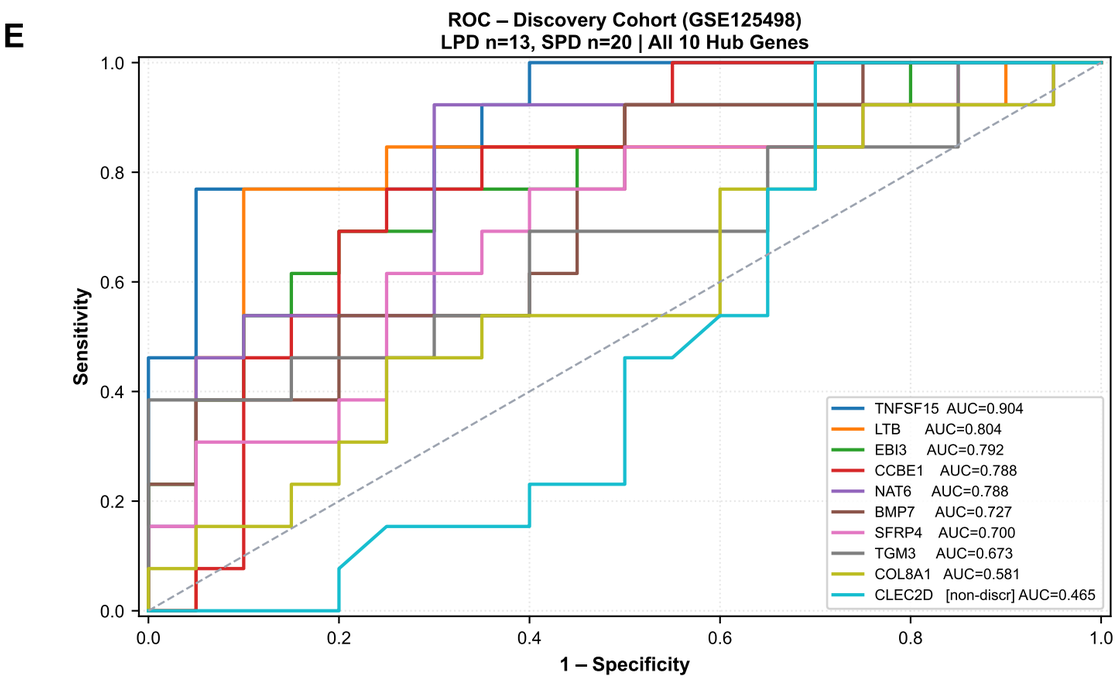


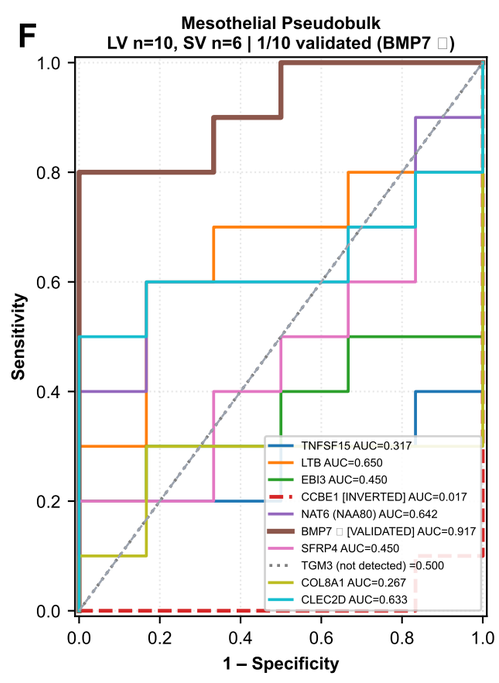


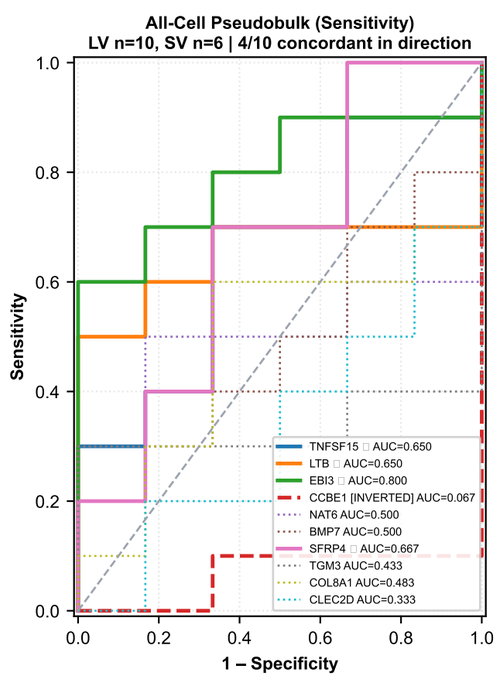




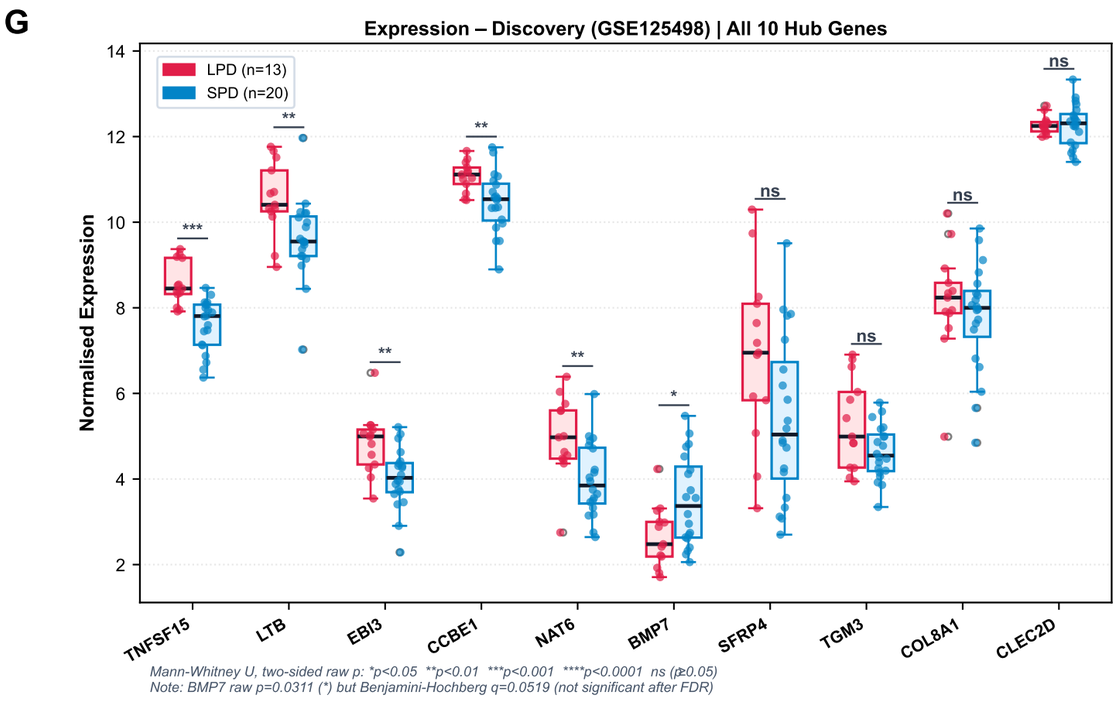




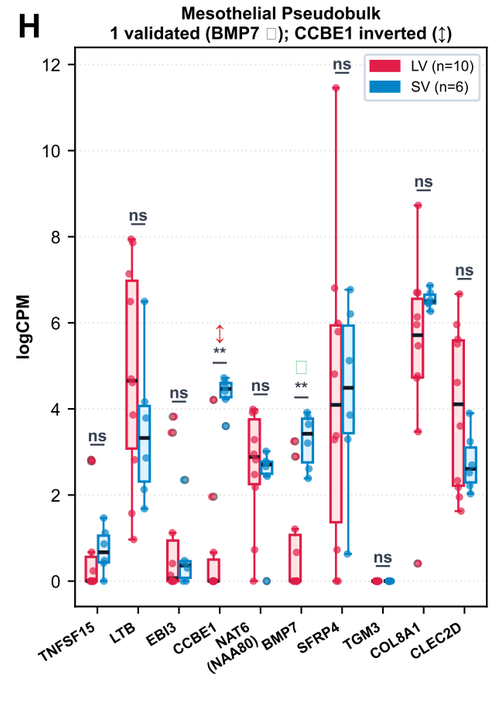


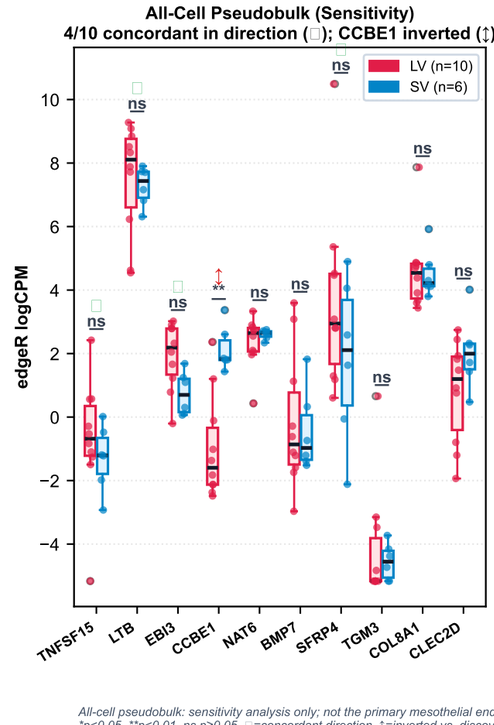


□ = same direction as discovery (LV > SV). CCBE1 significantly inverted in both pseudobulk types.
All-cell: 8/10 concordant in direction; Mesothelial: only BMP7 FDR-significant (q=0.025).



*Mann-Whitney U, two-sided raw p: *p<0.05 **p<0.01 ***p<0.001 ****p<0.0001 ns (p≥0.05)*
Note: BMP7 raw p=0.0311 (*) but Benjamini-Hochberg q=0.0519 (not significant after FDR)





**p<0.05 **p<0.01 ns p≥0.05 □=validated (BH q<0.05) ↕=inverted*

All-cell pseudobulk: sensitivity analysis only; not the primary mesothelial endpoint.
**p<0.05 **p<0.01 ns p≥0.05 □=concordant direction ↕=inverted vs. discovery*